

# AI-READABLE KNOWLEDGE BASE

## Department Reference Document

CSE - Data Science (CSD) | CSE - AI & Machine Learning (CSM)

---

### DEPARTMENT 1: CSE - DATA SCIENCE (CSD)

#### 1. Department Overview

The Department of Computer Science and Engineering - Data Science (CSD) offers a B.Tech program with a focus on data analysis, prediction, and the application of emerging technologies to solve real-world problems.

Data analysis and prediction rely increasingly on Data Science, and the speed with which computing technology continues to transform daily lives is truly astonishing. Specialists in this area are in demand worldwide because their applied skills can revolutionize businesses and organizations of all kinds.

The fast-changing technological scenario poses a continuous challenge of updating curriculum and faculty to provide state-of-the-art education to students. Both major and subtle changes have been made by the department to be better prepared for tomorrow.

#### Department Head

**Name:** Dr. P. Madhuravani

**Experience:** 15 years (Industrial and Teaching)

**Total Faculty:** 56 faculty members

**Faculty with Doctorates:** 9

#### Program Details

**Program:** B.Tech in Data Science

**Year Started:** 2020

**Initial Intake (2020):** 60

**Intake (2021 onwards):** 60

**EAPCET Code:** MLID

## 2. Vision and Mission

### Vision

Achieve global recognition through innovation, interdisciplinary research and excellence in Data Science. Empower students with professional competence to excel in industry, higher studies and entrepreneurship.

### Mission Statements

**M1:** To foster a culture of innovation and interdisciplinary research that addresses real-world challenges through the application of advanced Data Science techniques and emerging technologies.

**M2:** To equip students with strong analytical, technical and professional skills, enabling them to excel in industry, pursue higher education and engage in entrepreneurial ventures.

**M3:** To promote collaboration with academia, industry and research organizations globally, ensuring continuous learning, ethical practice and societal impact through Data Science.

## 3. Research Areas

Current areas of research in the department are broadly classified into the following categories:

- Multiple, Heterogeneous Data Sources
- Precious Data
- Inferring From Noisy and/or Incomplete Data
- Computing Systems for Data-Intensive Applications
- Automating Front-End Stages of the Data Life Cycle

## 4. Infrastructure and Laboratories

### Laboratory Facilities

The modern, air-conditioned laboratories serve the teaching and research needs of students and faculty.

Lab Name: Departmental Computer Laboratories

- Purpose: Teaching, research, and practical training in Data Science
- Facilities: All systems connected to the department LAN through fiber optic, which is part of the Institute LAN network
- Internet Connectivity: 300 Mbps link leased from Gateway and Vainavi
- Used By: B.Tech Data Science students and faculty

### Additional Facilities

- Central Computing Facility
- e-Classroom with multimedia facilities
- Internet access
- Centrally air-conditioned Seminar Hall / Auditorium

## 5. Teaching Methodology

The department employs a scientific teaching methodology to inculcate conceptual and technical skills in students. Methods include:

- Lectures
- Case Studies
- Seminars
- Group Discussions
- Project Works
- Assignments
- Quiz Programs

Teaching aids regularly used include LCD Projector, OHP, CD demonstrations, equipment demonstrations, models, and charts.

A systematic and well-planned approach to problem solving is emphasized in classrooms and laboratories through course projects and additional computer training beyond the scope of the syllabus.

### Mentorship

Around 18-20 students of each class are allocated to a mentor who maintains close rapport with each student regarding academic and personal matters.

## 6. Program Outcomes (POs)

A graduate of the Computer Science and Engineering - Data Science program will demonstrate:

**PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

**PO2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

**PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental factors.

**PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

**PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of limitations.

**PO6. The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

**PO7. Environment and Sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.

**PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

**PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports, design documentation, presentations, and clear instructions.

**PO11. Project Management and Finance:** Demonstrate knowledge of engineering and management principles and apply these to one's own work as a member and leader in a team to manage projects in multidisciplinary environments.

**PO12. Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

### Program Specific Outcomes (PSOs)

**PSO1:** Analyse and visualize data in the context of real-world problems, communicate findings, and interpret results using data analytics for decision making.

**PSO2:** Students of Data Science will explore interdisciplinary research and applications to solve industry problems and sustainable development goals.

## 7. Student Activities and Professional Bodies

### Professional Student Chapters

- IEEE (Institute of Electrical and Electronics Engineers)
- CSI (Computer Society of India)
- ISTE (Indian Society for Technical Education)

These chapters conduct events throughout the year, keeping students engaged with beyond-the-classroom activities.

### Notable Student Events

- MANTHAN 2021 HACKATHON (Virtual): Students of CSE-(AIML, CS, DS) participated in the AICTE-organized event on 09-10-2021.
- Design Thinking Workshop: Students of CSE-(AIML, CS, DS) attended the workshop on 09-10-2021 held at IIIT-Hyderabad.

### Faculty Activities

- Department of CSE (AIML) inaugurated Centre of Excellence For Artificial Intelligence & Machine Learning by Mr. Bapineedu Bollempalli, Senior Director, Virtusa, on 01st September 2021.
- Department of CSE-Cyber Security conducted a Faculty Training Programme on CYBER SECURITY by Z9 Labs Pvt Ltd, in association with CRCIDF.
- Department of CSE (Data Science) organized a Faculty Development Program on MACHINE LEARNING AND DATA SCIENCE FOR ENGINEERING APPLICATIONS in association with E&ICT Academy, NIT Warangal.

### Student Webinars and Guest Lectures

- TECH TALK on AI & ML Webinar (19th March) - Speaker: Fazeel Ahmed, Sr. Software Engineer, EPAM (CSE AIML department)
- Guest Lecture on The Future of Cyber Security (9th March) - Resource Person: Akhilesh Kanduri, CEO, Z9 Labs Pvt Ltd (CSE Cyber Security department)
- Webinar on Fundamentals of Cyber Safety & Security - Speaker: Mr. Tajwar Khan, CEO & Founder, Ethical Learner Pvt. Ltd. (CSE Cyber Security department)

## 8. Achievements

### Faculty Achievements

#### Life Member Award

**Recipient:** Mr. Para Upendar, Associate Professor, CSC

**Award:** Life Member of the society for promoting quality and standards of Science and Technical Education

**Date:** February 15, 2022

#### Young Researcher Award

**Recipient:** Mr. Para Upendar, Associate Professor, CSC

**Award:** Young Researcher Award at IJEMR-ELSEVIER SSRN RESEARCH AWARDS 2021

**Date:** February 15, 2022

## 9. Special Programs - SWAYAM

SWAYAM (Study Webs of Active Learning for Young Aspiring Minds) is a Government of India initiative providing opportunities for life-long learning. Learners can choose from hundreds of courses taught at university, college, or school level, offered by the best teachers in India and elsewhere.

### Key Features

- Students can transfer SWAYAM credits into their academic record.
- Available to anyone: working, non-working, in school, or out of school.
- Designed to achieve the three cardinal principles of Education Policy: Access, Equity and Quality.
- All courses are available free of cost to residents in India.
- More than 1,000 specially chosen faculty and teachers from across the country participated in preparing these courses.

### Course Format (4 Quadrants)

- Quadrant 1: Video Lecture
- Quadrant 2: Specially prepared reading material (downloadable/printable)
- Quadrant 3: Self-assessment tests through tests and quizzes
- Quadrant 4: Online discussion forum for clearing doubts

### National Coordinators

- NPTEL - Engineering
- UGC - Post-graduation education
- CEC - Under-graduate education
- NCERT & NIOS - School education
- IGNOU - Out-of-school students
- IIMB - Management studies

### NPTEL Coordinators (CSD Department)

| S.No. | Name           | Designation | Phone Number |
|-------|----------------|-------------|--------------|
| 1     | K. Pushpa Rani | Asst. Prof. | 7799375067   |
| 2     | G. Sowmya      | Asst. Prof. | 7780192601   |

## 10. Research Publications (CSD Department)

### Publications 2019-20

- Chandra Sekhar Reddy N, Vemuri P, Govardhan A. "An empirical study on support vector machines for intrusion detection." International Journal of Emerging Trends in Engineering Research (2019).
- Chandra Sekhar Reddy N, Vemuri P, Govardhan A. "An implementation of novel feature subset selection algorithm for IDS in mobile networks." International Journal of Advanced Trends in Computer Science and Engineering (2019).
- N. ChandraSekhar Reddy, M. Srinivasa Rao, Raja Rajeswari Thota, Y. Harini Reddy. "Vehicle Security System." International Journal of Innovative Technology and Exploring Engineering (IJITEE), ISSN: 2278-3075, Volume-9 Issue-3, January 2020.
- B. Madhuravani, N. Chandra Sekhar Reddy, K. Sai Prasad, B. Dhanalaxmi, V. Uma Maheswari. "Strong and Secure Mechanism for Data Storage in Cloud Environment." International Journal of Advanced Trends in Computer Science and Engineering, Volume 8, No.1.3, 2019, ISSN 2278-3091.
- Susmitha Valli Gogula, K. Shekar, P. Purushotham, Y. Teja Sai Bhavani. "Android based image processor for blind." International Journal of Innovative Technology and Exploring Engineering, Vol. 9(1), Nov/2019, PP: 1229-1232.
- Pushpa Rani K, Reddy M, Anjaneyulu B, Hari Chandana B. "A formal assessment paper on EDUSET-MBA universities custom search." International Journal of Innovative Technology and Exploring Engineering, Volume-9 Issue-1, November 2019, ISSN: 2278-3075.
- Pushpa Rani K, Lakshmi L, Teegala N, Dhanalaxmi. "An application for the development of smart library as an academic initiative." International Journal of Advanced Trends in Computer Science and Engineering, Volume No 8(1.3): PP:55-58, July 2019.
- K. Sai Prasad & Rajender Miryala. "Histopathological Image Classification Using Deep Learning Techniques." International Journal on Emerging Technologies 10(2): 467-473 (2019).
- K. Sai Prasad, N. Chandrasekhar Reddy, Puneeth B.N. "A Framework for Diagnosing Kidney Disease in Diabetes Patients Using Classification Algorithms." SN COMPUT. SCI. 1, 101.

- Sowmya G, Navya K, Divya Jyothi G. "Machine learning and mining for social media analytics." *Advances in Intelligent Systems and Computing* 978-981 (2019).
- N. Sirisha, K.V.D. Kiran. "An Efficient and Lightweight Security Scheme for Big Data." *International Journal on Emerging Technologies* 11(1): 414-420 (2020).
- Sk. Khaja Shareef, N. Shirisha. "Efficient Automation using DTMF." *International Journal of Recent Technology and Engineering (IJRTE)*, Volume-8 Issue-5, January 2020, 3965-3967.
- Venkata Siva Rao A, John Samuel, Divya Priya, B. Kavya Sree. "Inform Me - Family Location Tracker App." *International Journal of Innovative Technology and Exploring Engineering (IJITEE)*, ISSN: 2278-3075, Volume-8 Issue-12, October 2019, Page No. 4588-4590.
- Boggula Lakshmi, B. Madhuravani, B. Veda Vidya, C. Sowjanya. "SeSPHR: A Methodology for Secure Sharing of Personal Health Records in the Cloud." *International Journal IJRTE*, Volume 2, Issue 6, Print ISSN: 2395-1990, Online ISSN: 2394-4099, 2019, Page No.690-694.
- Boggula Lakshmi, B. Navya. "Energy Efficient Routing Mechanism for Harsh Environment in Wireless Sensor Networks." *International Journal IJETER*, Volume-7, Issue-9, E-ISSN: 2347-3983, 2019, Page No.234-238.
- R. Anusha, Boggula Lakshmi, T. Mounika, Spurthi Kankanala. "Deep Stock Prediction." *International Journal of Recent Technology and Engineering (IJRTE)*, ISSN: 2277-3878, Volume-8 Issue-4, November 2019.

## **Publications 2018-19**

- B. Madhuravani, B. Rama, N. Chandra Sekhar Reddy, B. Dhanalaxmi, V. UmaMaheswari. "A Mathematical Model of Integrated Chaotic Based Hash Function to Improve Randomization and Bit Variation Properties of Chaotic Maps." *J.Mech.Cont.& Math. Sci.*, Vol.-13, No.-5, November-December (2018), Pp: 225-240.
- B. Madhuravani, D. S. R. Murthy, S. V. Raju. "An Improved Wireless Node Neighbor Integrity Verification and Encryption Using Additive and Multiplicative Homomorphic Model." *Journal of Fundamental and Applied Sciences*, ISSN 1112-9867, 2018, 10(6S), 2911-2932.
- K.H. Vijaya Kumari, Susmitha Valli Gogula, R. Karthik. "Scalar vs Vector Color Image Processing: An Analysis." *Journal of Advanced Research in Dynamical and Control Systems (JARDCS)*, 2018, Issue 8.
- Susmitha Valli Gogula, Karthik Rajendra. "A Novel Approach For Noise Filtering From Diabetic Retinopathy Images Using Improved Pillar K-Means Algorithm." *Journal of Fundamental and Applied Sciences*, 2018, Issue 10(6s).
- Y. Md. Riyazuddin, G. Susmithavalli, G. Victor Daniel. "An Approach For The Implementation Of Double Guard Method For Detecting The IDS In Web Applications." *International Journal of Advanced Research in Management, Architecture, Technology and Engineering*, 2018, Vol 4(6).
- Susmitha Valli Gogula, Karthik Rajendra. "An Approach To Detect Bone Tumor Using Segmentation Technique." *International Journal of Pharmaceutical Sciences*, 2018, Vol 10(2).
- Karthik Rajendra, Susmitha Valli Gogula. "Design And Development Of An Automated Assistance Intelligent Programmed Tool For Medical Diagnosis (AAMD)." *International Journal of Pharmaceutical Sciences*, 2018, Vol 10(2).

- R. Karthik, Sankit R. Kassa, Susmitha Valli Gogula, B. Naresh. "Energy Efficient Reliable (2Nx1) Multiplexer Design using QCA Approach." The IIOAB Journal, 2018, Pp:113-117, Vol 9(1).
- G. Prabhakar Reddy, K. Sai Prasad, N. Chandra Shekar Reddy and R. Karthik. "Privacy Preserving and Data Publishing using Tuple Grouping Algorithm." Journal of Engineering and Applied Sciences, 13: 930-933, 2018.
- Kashi Sai Prasad, S. Pasupathy. "Real-time Data Streaming using Apache Spark on Fully Configured Hadoop Cluster." J.Mech.Cont.& Math. Sci., Vol.-13, No.-5, November-December (2018), Pages 164-176.
- Rama B, Sai Prasad K, Sultana A, Shekar K. "A geographical factor of interest recommended strategies in location based social networks." International Journal of Engineering and Technology (UAE), 7 (3.27), 2018, 32-35.
- Arul Dalton, Prabhakar Reddy G, Sai Prasad K. "Fuzzy logics associated with neural networks in intelligent control for better world." International Journal of Reasoning-based Intelligent Systems, 10, 322, 2018.
- P. R. M. Rao, K. L. Chugh, K. S. Prasad and K. Shekar. "Impact of Active Learning in Engineering Education." 2018 IEEE 6th International Conference on MOOCs, Innovation and Technology in Education (MITE), Hyderabad, India, 2018, pp. 107-114.
- Sowmya G, Divya Jyothi G, Shirisha N, Navya K. "Active learning strategies in engineering education." Journal of Advanced Research in Dynamical and Control Systems, Special Issue (03), 2018, 1253-1258.
- Sowmya G, Divya Jyothi, Shirisha N, Navya K, Padmaja B. "IoT based Smart Door Lock System." International Journal of Engineering and Technology (UAE), 7(3.6), 2018, 223-225.
- Pushpa Rani K, Roja G, Sabitha C, Dhana Lakshmi B, Sreeja. "A new approach for converse binary tree traversals." International Journal of Engineering and Technology (UAE), Volume 7, Issue 4, 2018.
- Roja G, Pushpa Rani K, Sabitha C, Dhanalaxmi B, Sreeja S. "IOT based novel session payment system." International Journal of Engineering and Technology (UAE), Volume 8, Issue 4, 2018.
- N. Thulasi Chitra, R. Anusha and Roja. "Locating a book in Library using Wi-Fi." International Journal of Recent Technology of Engineering, ISSN: 2277-3878, Volume-7 Issue-6, March 2019, Page No.: 114-117.
- N. Thulasi Chitra, K. Pushpa Rani and Roja. "A Credit Card Fake Detection System using Image Cryptography." International Journal of Recent Technology of Engineering, ISSN: 2277-3878, Volume-7 Issue-6, March 2019.
- B. Anandkumar, Chaitrali S. Dangare, A. Manusha Reddy, Y. Indu and B. Padmaja. "A survey on security in cloud computing." Journal of Advanced Research in Dynamical and Control Systems, Special Issue (11), 2018, 82-86.
- Sirisha N, Kiran K.V.D. "Stock exchange analysis using Hadoop user experience (Hue)." Proceedings of ICISS 2017, pp. 1141-1144, 2018.
- Sirisha N, Kiran K. "Authorization of data in Hadoop using Apache Sentry." International Journal of Engineering and Technology (UAE), 7 (3), 2018, pp. 234-236.
- Sirisha N, Kiran K.V.D, Karthik R. "Hadoop security challenges and its solution using KNOX." Indonesian Journal of Electrical Engineering and Computer Science, 12 (1), 2018, pp. 107-116.

## Publications 2017-18

- N. Chandra Sekhar Reddy. "Evaluation of PCA and K-means Algorithm for Efficient Intrusion Detection." IJAER Volume 12 Issue No.12, ISSN No:0973-4562, 2017.
- N. Chandra Sekhar Reddy. "A Roadmap to security in IoT." IJAER Volume 12 Issue No.19, ISSN No:0973-4562, 2017.
- N. Chandra Shekar Reddy. "An efficient anti phishing framework based on dynamic captcha." IJCET Volume 7, Issue 6, ISSN No:0976-6367, Nov 2016.
- N. Chandra Shekar Reddy. "Preventing Malware Proliferation in Large Scale Network Topology." IJSR Volume 5, Issue 11, Nov 2016.
- K. L. Chugh. "Analysis and implementation of Twitter Stream by Real-Time Traffic Detection." IJETR Volume 6, Issue 3, Nov 2016.
- K. L. Chugh. "A Schema for Optimizing Research Publications Metrics for National Ranking of Engineering Institutions in India." IJIET Volume 7, Issue 4, Dec 2016.
- G. Kiran Kumar. "An Empirical study of resource allocation in various aspects of cloud computing." JARDCS Volume 9, Issue 8, ISSN No:1943023X, 2017.
- G. Kiran Kumar. "A Cost Effective Multi-Cloud Data Hosting Scheme using Charm." IJTET Volume 7, Issue 4, e-ISSN:2278-621X, 2016.
- B. Rama. "A Novel Approach to Relational Collaborative Topic Regression via Consistently Incorporating Client." IJCTT Volume 44, Issue 1, ISSN No:2231-2803, 2017.
- B. Rama. "Clustering and summarization with timelines on Tweet Streams." IJETR Volume 6, Issue 3, Nov 2016.
- B. Rama. "On the Study of 3D fractals." IJSC, ISSN 1816 9503, 2016.
- B. Madhura Vani. "A hybrid parallel hash model based on multi-chaotic maps for mobile data security." JTAIT Volume 95, Issue No.3, Feb 2017.
- B. Madhuravani. "Efficient Authentication Protocol Using Cryptographic Hash Function & LSB Dynamic Hop Image Steganographic Technique for Secure Communication." ASCTET, ISBN: 978-1-78561-786-7, 2016.
- B. Madhuravani. "Analysis and Implementation of Twitter Stream by Real-Time Traffic Detection." IJETR Volume 6, Issue 3, ISSN: 2321-0869 (O) 2454-4698 (P), November 2016.
- B. Madhura Vani. "An Efficient Authentication Protocol to amplify collision resistance using Dynamic Cryptographic Hash Function & LSB Hop based Image Steganographic Technique." IJAER Volume 11, Number 7, ISSN 0973-4562, 2016.
- G. Prabhakar Reddy. "Clustering and Classification of Molecular data using Artificial Intelligence." Elsevier Journal ICAAMM, Volume 4, Issue 8, ISSN: 2277-9655, 2016.
- K. Sai Prasad. "Artificial Intelligence Approach For Classifying Molecular Dataset using Density based Technique with appropriate Euclidean Distance Measure." Elsevier, Volume 4, Issue 8, ISSN: 2277-9655, 2016.
- K. Sai Prasad. "Research Article Authentic-Time Metropolis-Scale Taxi Ridesharing." IJCR Volume 8, Issue 11, Nov 2016.
- K. Sai Prasad. "Diagnosing Venomous Facebook Applications." IJECS Volume 5, Issue 11, Nov 2016.
- N. Prameela. "Text Summarization using topic modeling and cluster based MapReduce framework." IJECS Volume 5, Issue 12, Dec 2016.
- K. Rajesh. "Routing for Mobile Ad Hoc and Wireless Sensor Network on location basis." IJECS Volume 5, Issue 12, Dec 2016.

- E. Amarnath Reddy. "Achieving High Quality Tweet Segmentation using the HybridSeg Framework." IJCTT Volume 41, Issue 1, ISSN:2231-2803, Nov 2016.
- Dr. G. Arul Dalton. "Assimilation of Various Databases into XML format." IJIET Volume 7, Issue 4, ISSN: 2319-1058, December 2016.
- T. Dharma Reddy. "A Python Tool for Outcome Based Education." IJETER Vol. 4, Issue 12, ISSN No:2454-6410, December 2016.
- T. Dharma Reddy. "Automated Assistance for Syllabus Understanding and Making (AASUM) Tool." IJSDR Volume 1, Issue 12, ISSN: 2455-2631, December 2016.
- P. Ram Mohan Rao. "Comparative Study of Privacy Preservation in Data Analytics." IJIET Volume 7, Issue 2, ISSN: 2319-1058, August 2016.
- P. Ram Mohan Rao. "Authorize Client Inspection and Data Dynamics for Secure Storage in Cloud Computing." IJCTT Volume 41, Number 1, ISSN: 2231-2803, Nov 2016.
- A. Venkata Siva Rao. "Shrewd street dividers driven by IOT technology." IJCIET, ISSN: 0976-6308, 2017.
- Dr. Illaiah Kavati. "Fast finger print retrieval using minutiae neighbor structure." Springer, ISBN 978-981-10-8569-7, 2016.
- Dr. Illaiah Kavati. "A clustering based indexing approach for biometric databases using decision level fusion." IJB, Vol. 9, No. 1, 2016.
- Dr. Illaiah Kavati. "Signboard text translator: A tourist guide." IJECE, ISSN 2088-8708, 2016.
- E. Amarnath Reddy. "A secure railway crossing system using IOT." IEEE, ISBN: 978-1-5090-5686-6, 2016.
- KVSN Rama Rao. "A smart heuristics scanner for an intrusion detection system using two stage machine learning technique." IJAIP, Vol.9 No.5/6, pp.519-529, 2017.
- B. Balakishna. "Drone control vehicle for prospective protected shipping." IJCET, Vol.8, Issue 8, ISSN 09766308, 2017.

## **Publications 2016-17**

- N. Chandra Sekhar Reddy. "An Instruction Detection system for secure distributed local action detection and retransmission of pack." IJSC, 1816-9503, 2016.
- B. Madhura Vani. "Strong authentication using dynamic hashing and steganography." ICCCA, ISBN: 978-1-4799-8890-7, May 2015.
- B. Madhura Vani. "On-line Engineering Education with Emphasis on Bloom's Taxonomy." JEET, Special Issue, ISSN 2394-1707, Feb 2016.
- B. Madhura Vani. "An Efficient Authentication To Amplify Collision Resistance." IJAER, Volume 11, ISSN: 0973-4562, 2015-2016.
- K. L. Chugh. "On-line Engineering Education with Emphasis on Bloom's Taxonomy." JEET, Special Issue, ISSN 2394-1707, Feb 2016.
- K. Archana. "To detect tracking system of Parking Vehicle by using Sensors and ArduinoYUN Device." IJIET, Volume 6, Issue 4, ISSN: 2319-1058, April 2016.
- K. Archana. "To Analyse and Detect the Corruption with Modernization Using Android Application." IJARCS SE, Volume 6, Issues 4, ISSN: 2277 128X, April 2016.
- G. Kiran Kumar. "College Bus Tracking Android Application using GPS." IJNIET, Volume 4, Issue 4, ISSN: 2319-6319, 2016.
- G. Kiran Kumar. "TDMA Scheduling With WSN." IJS, Volume 1, ISSN: 2379-1500, 2016.
- G. Kiran Kumar. "Mobile Monitoring And Tracking System." IJMETMR, Volume 3, ISSN: 2348-4845, 2015-2016.

- G. Kiran Kumar. "An Efficient method for Conducting online quiz using mobile application." IJIET, Volume 6, Issue 4, ISSN: 2319-1058, April 2016.
- Dr. G. Arul Dalton. "BER Performance And Antenna Selection For Best Transmission In Spatial Modulation." IJARMATE, Vol. 2, Special Issue 6, ISSN: 2454-9762, March 2016.
- T. Dharma Reddy. "Mobile Monitoring and Tracking System." IJMETMR, Volume 3, Issue 4, ISSN: 2348-4845, April 2016.
- P. Ram Mohan Rao. "Accessing And Monitoring Virtual Machines Using Software Interface." IJESC, Volume 6, ISSN: 2321-3361, 2015-2016.
- P. Ram Mohan Rao. "A Novel Approach for Personalized Web Search using Data Mining And Privacy Protection." IJSETR, Volume 5, Issue 1, ISSN 2319-8885, Jan 2016.
- Dr. G. Kiran Kumar. "A Novel Approach for Personalized Web Search using Data Mining And Privacy Protection." IJSETR, Volume 5, Issue 1, ISSN 2319-8885, Jan 2016.
- Venkata Siva Rao. "Digital Fuel Meter." IJMETMR, Volume 3, Issue, ISSN: 2348-4845, April 2016.
- Y. Madan Reddy. "Graphical Layout Food Ordering And Table Reservation System." IJEEST, Volume 2, ISSN: 2454-549X, 2015-2016.
- L. Lakshmi. "Blood Friend Using Mobile Application." IJCSE, Volume 8, ISSN: 0976-5166, 2015-2016.
- M. Jhansi. "Rescue System Identification by a Position Based Routing Protocol." IJARCSSE, Volume 6, ISSN: 2277-128X, 2015-2016.
- M. Jhansi. "Stimulating Routing Attacks in Adhoc Wireless Networks." ICCII, Volume 6, ISSN: 2229-3515, 2015-2016.
- N. Sirisha. "Cryptographic Techniques to Prevent Jamming Attacks." IJMETMR, Volume 3, ISSN: 2348-4845, 2015-2016.

---

## DEPARTMENT 2: CSE - AI & MACHINE LEARNING (CSM)

### 1. Department Overview

The Department of Computer Science and Engineering - Artificial Intelligence & Machine Learning (CSM) offers B.Tech programs with a focus on AI, ML, and intelligent systems development.

#### Department Head

**Name:** Dr. Kashi Sai Prasad

**Teaching Experience:** 10 years

#### CSE - AI & ML (AIML Specialization) Details

**Program:** B.Tech in CSE - AI & ML

**Year Started:** 2020

**Intake 2020:** 60

**Intake 2021:** 180

**Current Intake:** 180

**Total Faculty:** 35

**Faculty with Doctorates:** 5

### **CSE - AI & ML (CSM Specialization) Details**

**Program:** B.Tech in CSM

**Year Started:** 2020

**Intake 2020:** 60

**Intake 2021:** 180

**Intake 2024:** 420

**Total Faculty:** 31

**Faculty with Doctorates:** 4

### **Research Focus Areas (CSM)**

- Machine Learning
- Deep Learning
- Reinforcement Learning
- Robotics
- Natural Language Processing
- Computer Vision
- Augmented Reality / Virtual Reality (AR/VR)

## **2. Vision and Mission**

### **Vision**

To create competent professionals and researchers in Artificial Intelligence and Machine Learning with the skills and entrepreneurial mindset to drive innovation and address real-world challenges in the IT industry.

### **Mission Statements**

**M1:** Impart strong foundational knowledge in Artificial Intelligence and Machine Learning through quality teaching and innovative pedagogy.

**M2:** Equip students to apply AI and ML technologies for solving societal problems through experiential learning, research, and ethical practices.

**M3:** Encourage interdisciplinary teamwork, leadership, and entrepreneurial mindset to promote innovation with social responsibility.

## **3. Infrastructure and Laboratories**

### **Laboratory Facilities**

The modern, air-conditioned laboratories serve the teaching and research needs of students and faculty.

Lab Name: Departmental Computer Laboratories

- Purpose: Teaching, research, and practical training in AI and Machine Learning
- Facilities: All systems connected to the department LAN through fiber optic, part of the Institute LAN network
- Internet Connectivity: 300 Mbps link leased from Gateway
- Used By: B.Tech CSM/AIML students and faculty

### Additional Facilities

- Centre of Excellence for Artificial Intelligence & Machine Learning (inaugurated 01 September 2021 by Mr. Bapineedu Bollempalli, Senior Director, Virtusa)
- Central Computing Facility
- e-Classroom with multimedia facilities
- Internet access
- Centrally air-conditioned Seminar Hall / Auditorium

## 4. Teaching Methodology

The department employs a scientific teaching methodology to inculcate conceptual and technical skills in students. Methods include:

- Lectures
- Case Studies
- Seminars
- Group Discussions
- Project Works
- Assignments
- Quiz Programs

Teaching aids regularly used include LCD Projector, OHP, CD demonstrations, equipment demonstrations, models, and charts.

A systematic and well-planned approach to problem solving is emphasized in classrooms and laboratories through course projects and additional computer training beyond the scope of the syllabus.

### Mentorship

Around 18-20 students of each class are allocated to a mentor who maintains close rapport with each student regarding academic and personal matters.

## 5. Program Outcomes (POs)

A graduate of the Computer Science and Engineering - Cyber Security / AI-ML program will demonstrate:

**PO1. Engineering Knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

**PO2. Problem Analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

**PO3. Design/Development of Solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health, safety, cultural, societal, and environmental factors.

**PO4. Conduct Investigations of Complex Problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

**PO5. Modern Tool Usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of limitations.

**PO6. The Engineer and Society:** Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

**PO7. Environment and Sustainability:** Understand the impact of professional engineering solutions in societal and environmental contexts, and demonstrate knowledge of and need for sustainable development.

**PO8. Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

**PO9. Individual and Team Work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**PO10. Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, including writing effective reports, design documentation, presentations, and clear instructions.

**PO11. Project Management and Finance:** Demonstrate knowledge of engineering and management principles and apply these to one's own work as a member and leader in a team to manage projects in multidisciplinary environments.

**PO12. Life-long Learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

### Program Specific Outcomes (PSOs)

**PSO1:** Develop an in-depth knowledge and skill sets in human cognition, Artificial Intelligence, Machine Learning and data engineering for designing intelligent systems to address modern computing challenges.

**PSO2:** Evaluate, analyse and synthesize solutions for real-time problems in Artificial Intelligence and Machine Learning domain to conduct research in a wider theoretical and practical context.

## 6. Student Activities and Professional Bodies

### Professional Student Chapters

- IEEE (Institute of Electrical and Electronics Engineers)
- CSI (Computer Society of India)
- ACM (Association for Computing Machinery)
- ISTE (Indian Society for Technical Education)

## Notable Student Events

- MANTHAN 2021 HACKATHON (Virtual): Students of CSE-(AIML, CS, DS) participated in the AICTE-organized event on 09-10-2021.
- Design Thinking Workshop: Students of CSE-(AIML, CS, DS) attended the workshop on 09-10-2021 held at IIIT-Hyderabad.

## Faculty Activities

- Department of CSE (AIML) inaugurated Centre of Excellence For Artificial Intelligence & Machine Learning by Mr. Bapineedu Bollempalli, Senior Director, Virtusa, on 01st September 2021.
- Department of CSE (AIML, Data Science) organized Faculty Development Program on MACHINE LEARNING AND DATA SCIENCE FOR ENGINEERING APPLICATIONS in association with E&ICT Academy, NIT Warangal.

## Student Webinars and Guest Lectures

- TECH TALK on AI & ML Webinar - Speaker: Fazeel Ahmed, Sr. Software Engineer, EPAM (CSE AIML department)
- Guest Lecture on The Future of Cyber Security - Resource Person: Akhilesh Kanduri, CEO, Z9 Labs Pvt Ltd (CSE Cyber Security department)
- Webinar on Fundamentals of Cyber Safety & Security - Speaker: Mr. Tajwar Khan, CEO & Founder, Ethical Learner Pvt. Ltd. (CSE Cyber Security department)

## 7. Achievements

### Life Member Award

**Recipient:** Mr. Para Upendar, Associate Professor, CSC

**Award:** Life Member of the society for promoting quality and standards of Science and Technical Education

**Date:** February 15, 2022

### Young Researcher Award

**Recipient:** Mr. Para Upendar, Associate Professor, CSC

**Award:** Young Researcher Award at IJEMR-ELSEVIER SSRN RESEARCH AWARDS 2021

**Date:** February 15, 2022

## 8. Special Programs - SWAYAM

SWAYAM is a Government of India initiative providing opportunities for life-long learning. Learners can choose from hundreds of courses taught at university, college, or school level, offered by the best teachers in India and elsewhere.

### Key Features

- Students can transfer SWAYAM credits into their academic record.
- Available to anyone: working, non-working, in school, or out of school.

- Designed to achieve the three cardinal principles of Education Policy: Access, Equity and Quality.
- All courses are available free of cost to residents in India.
- More than 1,000 specially chosen faculty and teachers from across the country participated in preparing these courses.

### Course Format (4 Quadrants)

- Quadrant 1: Video Lecture
- Quadrant 2: Specially prepared reading material (downloadable/printable)
- Quadrant 3: Self-assessment tests through tests and quizzes
- Quadrant 4: Online discussion forum for clearing doubts

### NPTEL Coordinators (CSM Department)

| S.No. | Name           | Designation | Phone Number |
|-------|----------------|-------------|--------------|
| 1     | K. Pushpa Rani | Asst. Prof. | 7799375067   |
| 2     | G. Sowmya      | Asst. Prof. | 7780192601   |

## 9. Research Publications (CSM Department)

### Publications 2019-20

- Chandra Sekhar Reddy N, Vemuri P, Govardhan A. "An empirical study on support vector machines for intrusion detection." International Journal of Emerging Trends in Engineering Research (2019).
- Chandra Sekhar Reddy N, Vemuri P, Govardhan A. "An implementation of novel feature subset selection algorithm for IDS in mobile networks." International Journal of Advanced Trends in Computer Science and Engineering (2019).
- N. ChandraSekhar Reddy, M. Srinivasa Rao, Raja Rajeswari Thota, Y. Harini Reddy. "Vehicle Security System." IJITEE, ISSN: 2278-3075, Volume-9 Issue-3, January 2020.
- B. Madhuravani, N. Chandra Sekhar Reddy, K. Sai Prasad, B. Dhanalaxmi, V. Uma Maheswari. "Strong and Secure Mechanism for Data Storage in Cloud Environment." International Journal of Advanced Trends in Computer Science and Engineering, Volume 8, No.1.3, 2019, ISSN 2278-3091.
- Susmitha Valli Gogula, K. Shekar, P. Purushotham, Y. Teja Sai Bhavani. "Android based image processor for blind." International Journal of Innovative Technology and Exploring Engineering, Vol. 9(1), Nov/2019, PP: 1229-1232.
- Pushpa Rani K, Reddy M, Anjaneyulu B, Hari Chandana B. "A formal assessment paper on EDUSET-MBA universities custom search." IJITEE, Volume-9 Issue-1, November 2019, ISSN: 2278-3075.
- Pushpa Rani K, Lakshmi L, Teegala N, Dhanalaxmi. "An application for the development of smart library as an academic initiative." International Journal of Advanced Trends in Computer Science and Engineering, Volume No 8(1.3): PP:55-58, July 2019.
- K. Sai Prasad & Rajender Miryala. "Histopathological Image Classification Using Deep Learning Techniques." International Journal on Emerging Technologies 10(2): 467-473 (2019).

- K. Sai Prasad, N. Chandrasekhar Reddy, Puneeth B.N. "A Framework for Diagnosing Kidney Disease in Diabetes Patients Using Classification Algorithms." SN COMPUT. SCI. 1, 101.
- Sowmya G, Navya K, Divya Jyothi G. "Machine learning and mining for social media analytics." Advances in Intelligent Systems and Computing 978-981 (2019).
- N. Sirisha, K.V.D. Kiran. "An Efficient and Lightweight Security Scheme for Big Data." International Journal on Emerging Technologies 11(1): 414-420 (2020).
- Sk. Khaja Shareef, N. Shirisha. "Efficient Automation using DTMF." IJRTE, Volume-8 Issue-5, January 2020, 3965-3967.
- Venkata Siva Rao A, John Samuel, Divya Priya, B. Kavya Sree. "Inform Me - Family Location Tracker App." IJITEE, ISSN: 2278-3075, Volume-8 Issue-12, October 2019, Page No. 4588-4590.
- Boggula Lakshmi, B. Madhuravani, B. Veda Vidya, C. Sowjanya. "SeSPHR: A Methodology for Secure Sharing of Personal Health Records in the Cloud." IJRTE, Volume 2, Issue 6, 2019, Page No.690-694.
- Boggula Lakshmi, B. Navya. "Energy Efficient Routing Mechanism for Harsh Environment in Wireless Sensor Networks." IJETER, Volume-7, Issue-9, E-ISSN: 2347-3983, 2019, Page No.234-238.
- R. Anusha, Boggula Lakshmi, T. Mounika, Spurthi Kankanala. "Deep Stock Prediction." IJRTE, ISSN: 2277-3878, Volume-8 Issue-4, November 2019.

## Publications 2018-19

- B. Madhuravani, B. Rama, N. Chandra Sekhar Reddy, B. Dhanalaxmi, V. UmaMaheswari. "A Mathematical Model of Integrated Chaotic Based Hash Function..." J.Mech.Cont.& Math. Sci., Vol.-13, No.-5, November-December (2018), Pp: 225-240.
- B. Madhuravani, D. S. R. Murthy, S. V. Raju. "An Improved Wireless Node Neighbor Integrity Verification and Encryption Using Additive and Multiplicative Homomorphic Model." Journal of Fundamental and Applied Sciences, ISSN 1112-9867, 2018, 10(6S), 2911-2932.
- K.H. Vijaya Kumari, Susmitha Valli Gogula, R. Karthik. "Scalar vs Vector Color Image Processing: An Analysis." JARDCS, 2018, Issue 8.
- Susmitha Valli Gogula, Karthik Rajendra. "A Novel Approach For Noise Filtering From Diabetic Retinopathy Images Using Improved Pillar K-Means Algorithm." Journal of Fundamental and Applied Sciences, 2018, Issue 10(6s).
- Y. Md. Riyazuddin, G. Susmithavalli, G. Victor Daniel. "An Approach For The Implementation Of Double Guard Method For Detecting The IDS In Web Applications." IJARMAE, 2018, Vol 4(6).
- Susmitha Valli Gogula, Karthik Rajendra. "An Approach To Detect Bone Tumor Using Segmentation Technique." International Journal of Pharmaceutical Sciences, 2018, Vol 10(2).
- Karthik Rajendra, Susmitha Valli Gogula. "Design And Development Of AAMD." International Journal of Pharmaceutical Sciences, 2018, Vol 10(2).
- R. Karthik, Sankit R. Kassa, Susmitha Valli Gogula, B. Naresh. "Energy Efficient Reliable (2Nx1) Multiplexer Design using QCA Approach." IIOAB Journal, 2018, Vol 9(1), Pp:113-117.
- G. Prabhakar Reddy, K. Sai Prasad, N. Chandra Shekar Reddy and R. Karthik. "Privacy Preserving and Data Publishing using Tuple Grouping Algorithm." Journal of Engineering and Applied Sciences, 13: 930-933, 2018.

- Kashi Sai Prasad, S. Pasupathy. "Real-time Data Streaming using Apache Spark on Fully Configured Hadoop Cluster." J.Mech.Cont.& Math. Sci., Vol.-13, No.-5, 2018, Pages 164-176.
- Rama B, Sai Prasad K, Sultana A, Shekar K. "A geographical factor of interest recommended strategies in location based social networks." IJET (UAE), 7 (3.27), 2018, 32-35.
- Arul Dalton, Prabhakar Reddy G, Sai Prasad K. "Fuzzy logics associated with neural networks in intelligent control for better world." International Journal of Reasoning-based Intelligent Systems, 10, 322, 2018.
- P. R. M. Rao, K. L. Chugh, K. S. Prasad and K. Shekar. "Impact of Active Learning in Engineering Education." IEEE MITE 2018, Hyderabad, India, pp. 107-114.
- Sowmya G, Divya Jyothi G, Shirisha N, Navya K. "Active learning strategies in engineering education." JARDCS, Special Issue (03), 2018, 1253-1258.
- Sowmya G, Divya Jyothi, Shirisha N, Navya K, Padmaja B. "IoT based Smart Door Lock System." IJET (UAE), 7(3.6), 2018, 223-225.
- Pushpa Rani K, Roja G, Sabitha C, Dhana Lakshmi B, Sreeja. "A new approach for converse binary tree traversals." IJET (UAE), Volume 7, Issue 4, 2018.
- Roja G, Pushpa Rani K, Sabitha C, Dhanalaxmi B, Sreeja S. "IOT based novel session payment system." IJET (UAE), Volume 8, Issue 4, 2018.
- N. Thulasi Chitra, R. Anusha and Roja. "Locating a book in Library using Wi-Fi." IJRTE, ISSN: 2277-3878, Volume-7 Issue-6, March 2019, Page No.: 114-117.
- N. Thulasi Chitra, K. Pushpa Rani and Roja. "A Credit Card Fake Detection System using Image Cryptography." IJRTE, ISSN: 2277-3878, Volume-7 Issue-6, March 2019.
- B. Anandkumar et al. "A survey on security in cloud computing." JARDCS, Special Issue (11), 2018, 82-86.
- Sirisha N, Kiran K.V.D. "Stock exchange analysis using Hadoop user experience (Hue)." ICISS 2017, pp. 1141-1144, 2018.
- Sirisha N, Kiran K. "Authorization of data in Hadoop using Apache Sentry." IJET (UAE), 7 (3), 2018, pp. 234-236.
- Sirisha N, Kiran K.V.D, Karthik R. "Hadoop security challenges and its solution using KNOX." Indonesian Journal of Electrical Engineering and Computer Science, 12 (1), 2018, pp. 107-116.

## **Publications 2017-18**

(Selected key publications from the CSM department faculty in 2017-18)

- K. Srinivas Rao. "Fast finger print retrieval using minutiae neighbor structure." Springer, Volume 705, ISBN 978-981-10-8569-7, 2018.
- K. Srinivas Rao. "Secure privacy preserving of personal health records using attribute based encryption in cloud computing." AICC, Volume 815, ISSN 2194-5357, 2018.
- K. L. Chugh. "Blended learning using E-shikshak." ICISS, ISBN: 978-1-5386-1959-9, June 2018.
- K. L. Chugh. "Quality research publications - A key differentiator." IJMET, Volume 8, Issue 6, 2018.
- K. L. Chugh. "Methodology for Self-Assessment in the context of National Rankings of Engineering Institutions in India." IJCIET, Volume 8, Issue 5, ISSN No:0976-6308, 2018.
- N. Chandra Sekhar Reddy. "An approach to enhance security using Reversible Texture synthesis process." ICECA, ISSN No:133534, 2017.

- N. Chandra Sekhar Reddy. "An empirical study on Feature Extraction techniques for instruction detection system." JARDCS, Volume 9, Issue 2, 2017.
- N. Chandra Sekhar Reddy. "DetectsStress: A novel stress detection system based on smart phone wireless physical activity tracker." AICC, ISBN 978-981-13-1580-0, 2018.
- G. Kiran Kumar. "Spatial Co-location Pattern mining using Delaunay Triangulation." Springer, ISBN: 978-981-10-8569-7, May 2018.
- B. Rama. "The Development of 3D Fractal Shapes for Mandelbrot and Julia Sets." IJPAM, Volume 118, Issue 15, ISSN No:1314-3395, 2018.
- T. Dharma Reddy. "Automated Assistance for Foundation Engineers." IJARCS, Volume 9, Issue 2, ISSN No:0976-5697, 2018.
- T. Dharma Reddy. "Automated Course Outcomes Assessment for MCQS." IJARCS, Volume 8, No.9, ISSN No.0976-5697, Nov 2017.
- G. Susmitha Valli. "Bioinformatics and image processing - Detection of plant diseases." Springer, ISBN: 978-981-13-1580-0, Volume 815, 2018.
- G. Prabhakar Reddy. "Featured Based Pattern Analysis using Machine Learning and Artificial Intelligence Techniques for Multiple Featured Dataset." Elsevier ICAAMM-2016, Volume 4, Dec 2017.
- K. Sai Prasad. "Analyzing and Prediction of Academic Performance of Students using Data Mining Techniques." JARDCS, Issue 7 Special, ISSN No:1943-023X, 2018.
- P. Ram Mohan Rao. "Privacy preservation in big data analytics: a survey." Springer, ISSN 2196-1115, Sept 2018.
- Dr. Illaiah Kavati. "Score based indexing and retrieval technique for biometric databases." IJPRAI, Vol. 31, No. 06, 1756009, 2017.
- Dr. Illaiah Kavati. "Hierarchical decomposition of extended triangulation for fingerprint indexing." Springer, ISBN 978-3-319-57660-2, 2017.
- Dr. Illaiah Kavati. "An efficient score based indexing technique for fast palm print retrieval." Springer, ISBN 978-3-319-57660-2, 2017.

## **Publications 2016-17**

- N. Chandra Sekhar Reddy. "Evaluation of PCA and K-means Algorithm for Efficient Intrusion Detection." IJAER, Volume 12, Issue 12, ISSN No:0973-4562, 2017.
- N. Chandra Sekhar Reddy. "A Roadmap to security in IoT." IJAER, Volume 12, Issue 19, ISSN No:0973-4562, 2017.
- K. L. Chugh. "Analysis and implementation of Twitter Stream by Real-Time Traffic Detection." IJETR, Volume 6, Issue 3, Nov 2016.
- G. Kiran Kumar. "An Empirical study of resource allocation in various aspects of cloud computing." JARDCS, Volume 9, Issue 8, ISSN No:1943023X, 2017.
- B. Rama. "A Novel Approach to Relational Collaborative Topic Regression via Consistently Incorporating Client." IJCTT, Volume 44, Issue 1, ISSN No:2231-2803, 2017.
- B. Madhura Vani. "A hybrid parallel hash model based on multi-chaotic maps for mobile data security." JTAIT, Volume 95, Issue 3, Feb 2017.
- K. Sai Prasad. "Artificial Intelligence Approach For Classifying Molecular Dataset using Density based Technique." Elsevier, Volume 4, Issue 8, ISSN: 2277-9655, 2016.
- E. Amarnath Reddy. "Achieving High Quality Tweet Segmentation using the HybridSeg Framework." IJCTT, Volume 41, Issue 1, ISSN:2231-2803, Nov 2016.

- Dr. G. Arul Dalton. "Assimilation of Various Databases into XML format." IJiet, Volume 7, Issue 4, ISSN: 2319-1058, December 2016.
- P. Ram Mohan Rao. "Comparative Study of Privacy Preservation in Data Analytics." IJiet, Volume 7, Issue 2, ISSN: 2319-1058, August 2016.
- Dr. Illaiah Kavati. "Fast finger print retrieval using minutiae neighbor structure." Springer, ISBN 978-981-10-8569-7, 2016.
- Dr. Illaiah Kavati. "A clustering based indexing approach for biometric databases using decision level fusion." IJB, Vol. 9, No. 1, 2016.
- E. Amarnath Reddy. "A secure railway crossing system using IOT." IEEE, ISBN: 978-1-5090-5686-6, 2016.
- KVSN Rama Rao. "A smart heuristics scanner for an intrusion detection system using two stage machine learning technique." IJAIP, Vol.9 No.5/6, pp.519-529, 2017.
- B. Balakishna. "Drone control vehicle for prospective protected shipping." IJCET, Vol.8, Issue 8, ISSN 09766308, 2017.